



# Arosh Wijesinghe

Artificial Intelligence Undergraduate | AI/ML Internship Candidate

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[LinkedIn](#) | [GitHub](#) | [Portfolio](#)

## PROFILE

Third-year BSc (Hons) in Artificial Intelligence undergraduate at the University of Moratuwa with hands-on exposure to machine learning, LLM-based applications, RAG, computer vision, predictive analytics, and full-stack AI systems. I have project-level familiarity with Python, FastAPI, React/Next.js, Supabase/PostgreSQL, PyTorch, TensorFlow, Scikit-learn, and Hugging Face Transformers while continuously learning and improving. Seeking an AI/ML internship to contribute to practical AI product development and grow through industry mentorship.

## EDUCATION

### BSc (Hons) in Artificial Intelligence

2024 - 2028

University of Moratuwa | Second-year undergraduate

### GCE Advanced Level

Sri Lanka

Poramadulla Central College | AAB - Combined Mathematics, ICT, Physics | Z-score: +2.0535

## KEY PROJECTS

### PredictiX - AI-Powered Predictive Maintenance & Smart Ticket Categorization

Group Project | Ongoing

- Contributing to an AI-powered warehouse asset management platform with predictive maintenance, asset health scoring, smart ticket categorization, help desk workflows, dashboards, and analytical reporting.
- Worked on LLM-based chatbot handling and RAG-powered query answering to retrieve asset status, ticket progress, failure risk, and report insights from stored system data.
- Handled Supabase/PostgreSQL database work for users, assets, tickets, maintenance logs, predictive outputs, and system records; contributed to ticket section UI and help desk workflow.
- Working on predictive maintenance models, including survival analysis, to estimate asset failure risk and maintenance timelines; project is planned for research paper development.

**Technologies:** Python, FastAPI, Supabase, PostgreSQL, RAG, LLMs, React, Next.js, Tailwind CSS, Scikit-learn, XGBoost, LightGBM, Pandas, NumPy

### Smart Chair - AI & IoT Posture Analysis System

CM-1900 Intelligent Machine Project

- Built a smart chair prototype to analyze sitting posture and help identify posture habits that may cause discomfort, back pain, or long-term health issues.
- Implemented a four-step analysis concept: seat weight distribution, lumbar support pressure analysis, hip-to-upper-body angle checking, and side elevation checking.
- Integrated load cells, FSR 402 pressure sensors, ESP camera, and HX711 amplifier modules with ESP32-based hardware.
- Used OpenCV and MediaPipe for camera-based posture analysis, with Flask support for system interaction and data handling.

**Technologies:** ESP32, Flask, OpenCV, MediaPipe, Load Cells, FSR 402 Pressure Sensors, ESP Camera, HX711

Link: [LinkedIn Project Post](#)

## PROJECTS & PRACTICAL WORK

### River Water Level Regression - Data Science Forecasting Project

Academic ML Project

- Built regression-based models to estimate river water level using previous water level and rainfall-related features.
- Handled data cleaning, missing value treatment, exploratory analysis, model training, validation, residual diagnostics, and evaluation using MAE, MSE, RMSE, and R-squared.

**Technologies:** Python, Pandas, NumPy, Scikit-learn, Statsmodels, Matplotlib, Seaborn

### Tummy Boy - Fitness Tracking Mobile Application

Mobile App

- Developed a mobile fitness application for workout tracking, pushup counting, progress monitoring, GPS-based running distance ideas, and personalized workout suggestions.

**Technologies:** Flutter, Dart, Mobile App Development, Fitness Analytics Concept

Link: [GitHub](#)

PROJECTS & PRACTICAL WORK (CONTINUED)

- Money Balancing App - Finance & Productivity Mobile Application

Mobile App

- Developed a mobile application to record income, expenses, and money-related activities with a focus on budgeting and personal finance habits.

Technologies: Flutter, Dart, Mobile App Development

Link: [GitHub](#)
- Portfolio Website

Web Project

- Built and deployed a responsive portfolio website with dark/light mode UI, glassmorphism styling, animations, project showcases, and professional links.

Technologies: React, Tailwind CSS, Framer Motion, Vercel

Link: [Live Portfolio](#)
- BulkThreads - Group Buying Platform for Premium Clothing

Web Platform

- Built a modern group-buying platform for premium clothing where users join campaigns to reach a minimum quantity target and unlock wholesale pricing.
  - Implemented campaign-focused user flows and backend integration using Next.js 16, FastAPI, and Supabase with PostgreSQL and Auth.

Technologies: Next.js 16, FastAPI, Python, Supabase, PostgreSQL, Auth

Link: [GitHub](#)

TECHNICAL EXPOSURE & TOOLS

Tools listed below are based on coursework and project exposure; I am continuously learning and improving them through practical work.

|                   |                                                                                                              |
|-------------------|--------------------------------------------------------------------------------------------------------------|
| Programming       | Python, Java, C, Prolog, HTML, CSS, PHP, SQL                                                                 |
| AI / ML           | PyTorch, TensorFlow, Scikit-learn, LightGBM, XGBoost, BERT, Hugging Face Transformers, OpenCV, Pandas, NumPy |
| LLMs / GenAI      | RAG, LLaMA, GPT, Gemini, DeepSeek, Prompt Engineering, AI-assisted Development                               |
| Web / Backend     | React, Next.js, Tailwind CSS, FastAPI, Supabase, REST APIs                                                   |
| Databases         | PostgreSQL, MySQL, MongoDB, Supabase                                                                         |
| Tools / Platforms | Git, GitHub, VS Code, Jupyter Notebook, Docker, CI/CD Pipelines, Vercel, Firebase, Google Colab, Kaggle      |

CERTIFICATIONS

- Intro to Machine Learning - Kaggle, 2026
- Discovering Generative AI - Udacity, 2025
- Python - CODL, University of Moratuwa, 2024
- Web Design - CODL, University of Moratuwa

Certificate Folder: [View All Certificates](#)

ADDITIONAL INFORMATION

Extra-Curricular

School Prefect | Air Force Cadet

Languages

Sinhala - Native | English - Working Proficiency

Additional Initiative

Advance Loop Solutions - Small software initiative started with a friend; completed one small project as a learning/business experiment and is not currently active full-time.

REFERENCES

Mrs. K A Dilini T Kulawansa  
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